

Quantum Relay-Assisted Free-Space Optical Communication

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Abstract—Relay-based quantum communication utilizes intermediate nodes to extend the transmission distance of quantum signals and mitigate the masking effect caused by obstacles or turbulence in free-space optical (FSO) channels. In this study, we investigate a three-node quantum communication system, where Alice, the transmitter, sends quantum-encoded binary information to Bob, the ultimate receiver, via an intermediate relay node, Charlie. By introducing Charlie, the end-to-end transmission is divided into two shorter FSO links, improving signal stability and reliability in the presence of severe atmospheric turbulence and noise. This work proposes a quantum relaying method specifically designed for FSO links and examines the impact of two-hop relaying in terms of the probability of error. The Helstrom error bound is derived to assess the system's detection limits, while density operator calculations are performed to characterize the quantum relaying setup. In addition, a closed-form expression is obtained for key system performance metrics. The findings offer valuable insights into optimizing quantum relaying strategies for long-distance quantum communication over FSO channels.

Index Terms—Quantum communication, relaying, free-space optical (FSO) communication.

I. INTRODUCTION

QUANTUM communication systems leverage the principles of quantum mechanics to enable secure and efficient information transfer [1], [2], [3], [4]. Unlike classical communication, which relies on conventional bits (0 s and 1 s), quantum communication utilizes quantum bits or qubits, which can exist in superposition states. A key advantage of this technology is its inherent security, achieved through Quantum Key Distribution (QKD), where any attempt at eavesdropping alters the quantum state, making detection possible [3], [5], [6]. Long-distance quantum key transmission has been explored in [7], [8], while free-space optical quantum Binary Phase Shift Keying (BPSK) communications in turbulent channels are studied in [9]. Quantum entanglement further enables long-distance communication by creating correlated states between distant parties [10].

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Moreover, the exploration of signaling-based modulation for continuous-variable QKD in the context of indoor visible light communications is presented in [11]. In [12], Terahertz QKD for multiple input and multiple output (MIMO) systems is discussed, while [13] addresses transmission probabilities in QKD systems over free-space optical (FSO) links. The proposal for the design and analysis of an octaphase shift keying-based QKD system is made in [14], and the quantum advantages of binary discrete modulations for space channels are explored in [15]. The evolution and networking perspectives of QKD are thoroughly examined in [16], [17], [18], and eavesdropping detection in BB84 protocols is proposed in [19]. A scheme for fault tolerantly dealing with losses (or other “leakage” errors) in cluster state computation with some loss is proposed in [20]. A review, focuses on continuous-variable quantum information processes that rely on any combination of Gaussian states, Gaussian operations, and Gaussian measurements is presented in [21]. Quantum-mechanical communication theory is presented in [22].

Quantum key discrimination is thoroughly analyzed in [23]. Eliminating detector side-channel attacks has long been a challenging issue in quantum cryptography. A measurement-device-independent quantum key distribution is proposed in [24]. One modulation scheme, Quadrature Phase Shift Keying (QPSK), encodes information in four distinct coherent states, each represented by different phases in the complex plane [25]. In contrast to BPSK, where states are easily distinguishable, the non-orthogonality of QPSK states complicates optimal state discrimination. To achieve minimal detection error, advanced quantum measurement techniques are required, with quantum measurement theory providing the framework for utilizing optimal Positive Operator-Valued Measures (POVMs) [26], [27], [28], [29], [30], [31], [32]. Performance analysis of spectral-amplitude-coding optical code division multiple access (SAC-OCMA) using Identity Column Shift Matrix (ICSM) code is conducted for FSO transmission systems to evaluate system efficiency, BER, and resilience under atmospheric turbulence in [33]. The efficiency of FSO communication is enhanced using FSO and dense wavelength division multiplexing FSO and a hybrid approach that combines on-off keying (OOK) modulation, M-ary digital pulse position modulation (M-ary DPPM) and M-pulse amplitude and position modulation (M-PAPM) in [34], [35]. A Design of rate adaptation for relay-assisted satellite hybrid FSO/RF systems and Outage Probability and average BER of UAV-assisted dual-hop FSO communication are discussed in [36], [37], [38].

However, challenges such as photon loss, decoherence, and the need for quantum repeaters limit practical implementation [39]. Research efforts are focused on quantum relays, error correction, and satellite-based quantum networks to overcome these hurdles, paving the way for the development of a global Quantum Internet with ultra-secure data transmission [40], [41], [42], [43]. One of the major challenges in quantum communication is overcoming distance limitations due to photon loss and noise in optical fibers or free-space channels. Relay-based transmission is a key approach to extend the range of quantum communication.

A. Motivation

Quantum relays play a crucial role in free-space optical (FSO) quantum communication, particularly when there are obstacles or the transmission path is blocked. In classical and quantum FSO communication, line-of-sight (LOS) is essential for direct transmission. If a building, tree, or other obstacle blocks the direct path, quantum signals cannot reach the receiver. A relay station positioned strategically can redirect or retransmit the quantum signal, ensuring that it reaches the intended recipient. Another advantage of relay is to use very low power signals over short FSO Links [44], [45], [46]. Since quantum communication operates with a small number of photons or weak coherent state pulses, meaning that signal power is inherently low [47]. Over long distances, photon loss increases significantly due to link length dependence of the turbulence severity in the FSO links. The information reliability can be improved by placing relays in short-segment FSO links [48]. Another problems are severe Atmospheric conditions (like fog, rain, turbulence), that cause significant scattering and absorption of quantum signals, especially over long FSO links [49]. Instead of a single long-range FSO link, Quantum relays can divide the link into multiple shorter segments [50]. Each segment can be optimized for minimal atmospheric interference. Adaptive optics can be used to improve alignment and photon detection. Moreover, a direct quantum link is highly fragile, and any environmental change can lead to complete loss of communication [47]. By incorporating multiple quantum relay nodes, the network dynamically reroutes signals in case of a failure or obstruction. This approach makes quantum networks more scalable and reliable [51]. In real-world urban environments, physical obstructions such as buildings and trees can severely degrade FSO link performance, while satellite-based quantum communications suffer from significant photon loss and decoherence over long distances. Introducing quantum relays to subdivide these long links into shorter, more robust segments not only improves signal stability but also enhances security by mitigating cumulative atmospheric impairments. Therefore, this relaying strategy is essential for realizing practical, scalable quantum communication networks.

B. Novelty and Contributions

Relay-based quantum communication leverages intermediate nodes (relays) to extend the transmission distance and overcome channel impairments. In this study, we analyze a three-node quantum communication system where Alice, the transmitter,

sends quantum-encoded binary information to Bob, the ultimate receiver, via an intermediate relay node, Charlie. By dividing the long FSO channel into two shorter links, our approach significantly mitigates the detrimental effects of turbulence and noise, thereby enhancing signal stability and reliability. Our main contributions are as follows:

- 1) We propose a novel quantum relaying method for FSO links that efficiently converts classical bits to quantum states, enabling robust decode-and-forward (DF) relaying.
- 2) We analyze the overall impact of the two-hop relaying scheme by evaluating the end-to-end error probability, demonstrating improved performance over direct transmission.
- 3) We derive the Helstrom error bound for the system, which sets the fundamental limit on the minimum detection error when distinguishing non-orthogonal coherent states under realistic channel impairments.
- 4) We perform a detailed density operator analysis for the quantum relaying setup, incorporating the effects of Gamma–Gamma turbulence and additive Gaussian noise.
- 5) We obtain a closed-form expression for the error rate under Helstrom detection, rigorously accounting for amplitude fading and noise, and validate these results through extensive Monte Carlo simulations.

These contributions not only provide a rigorous theoretical framework but also demonstrate, both analytically and experimentally, that two-hop relaying can substantially reduce photon loss and decoherence in long-distance FSO channels, thus laying a robust foundation for scalable quantum communication networks.

II. QUANTUM COMMUNICATION: BASIC CONCEPTS

In this section, we will review few basic concepts of quantum communication.

A. Quantum States and Hilbert Spaces

In quantum mechanics, the state of a single optical mode is described by $|\psi\rangle$ in a (potentially infinite-dimensional) Hilbert space $\mathcal{H}$. More generally, one may use a *density operator* ρ , which is a positive semi-definite operator with unit trace, to represent possible statistical mixtures or interactions with the environment [4], [52]. Pure states correspond to rank-1 projectors: $\rho = |\psi\rangle\langle\psi|$.

B. Fock (Number) Basis

For a single-mode optical field, a convenient orthonormal basis is the *number basis*, $\{|n\rangle\}_{n=0}^{\infty}$, where $|n\rangle$ indicates exactly n photons [52], [53]. Any pure state can be expanded as

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle, \quad \text{with} \quad \sum_{n=0}^{\infty} |c_n|^2 = 1.$$

C. Non-Orthogonality and the No-Cloning Principle

Two states $|\psi\rangle$ and $|\phi\rangle$ may be *non-orthogonal*, i.e. $\langle\psi|\phi\rangle \neq 0$. In such cases, one cannot perfectly discriminate them in a

single-shot measurement. Moreover, the *no-cloning theorem* forbids making a perfect copy of an unknown quantum state, which contrasts with classical repeater strategies that copy signals freely [54].

D. Coherent States

A *coherent state* $|\alpha\rangle$, with $\alpha \in \mathbb{C}$, is defined via the displacement operator $D(\alpha)$ acting on the vacuum [52]:

$$|\alpha\rangle = D(\alpha) |0\rangle = \exp(\alpha a^\dagger - \alpha^* a) |0\rangle,$$

where $a, a^\dagger$ are annihilation/creation operators. In the Fock basis,

$$|\alpha\rangle = \exp\left[-\frac{1}{2}|\alpha|^2\right] \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (1)$$

Crucially, coherent states are non-orthogonal:

$$\langle\alpha|\beta\rangle = \exp\left[-\frac{1}{2}(|\alpha|^2 + |\beta|^2) + \alpha^* \beta\right]. \quad (2)$$

Hence, perfect discrimination between two coherent states is impossible if they are not infinitely separated in phase space.

E. Binary Communication With Non-Orthogonal States

In many quantum-communication or quantum-key distribution (QKD) protocols [55], *binary* states are employed. A classic example is:

$$|\psi_0\rangle = |+\alpha\rangle, \quad |\psi_1\rangle = |-\alpha\rangle,$$

where the overlap $\langle+\alpha|-\alpha\rangle \neq 0$. This overlap forces a fundamental detection error.

F. Quantum Measurements and Helstrom's Minimum-Error Bound

A general quantum measurement is described by a set of positive operators $\{M_i\}$, called a Positive Operator-Valued Measure (POVM), satisfying $\sum_i M_i = I$ [31], [32]. For two candidate states ρ_0 and ρ_1 with prior probabilities p_0 and p_1 , one wants to minimize the average error in deciding which state arrived.

G. Helstrom Theorem

Helstrom showed that for binary states the *minimum* probability of error is [56], [57]

$$P_e^{\min} = \frac{1}{2} \left[1 - \text{Tr} |p_1 \rho_1 - p_0 \rho_0| \right], \quad (3)$$

where $|X| = \sqrt{X^\dagger X}$ is the positive operator in the polar decomposition of X . When the states are pure and have equal priors, $\rho_0 = |\psi_0\rangle\langle\psi_0|$ and $\rho_1 = |\psi_1\rangle\langle\psi_1|$, with overlap $|\langle\psi_0|\psi_1\rangle|^2 = c^2$, this simplifies to

$$P_e^{\min} = \frac{1}{2} \left(1 - \sqrt{1 - c^2} \right). \quad (4)$$

For $|+\alpha\rangle$ vs. $|-\alpha\rangle$, one has $c = \exp(-2|\alpha|^2)$, leading to the well-known expression

$$P_e^{\min} = \frac{1}{2} \left(1 - \sqrt{1 - \exp[-4|\alpha|^2]} \right). \quad (5)$$

H. Photon Counting and Shot Noise

At the physical detection level, photons arrive discretely, following Poisson statistics (shot noise) [58]. In many classical FSO scenarios with high flux, one approximates this as Gaussian noise. In truly *quantum* low-photon regimes, the discrete nature dominates, hence one must carefully treat the probability of detecting n photons, especially when states are weak coherent or single photons. Non-orthogonality prevents perfect discrimination, and Helstrom's bound in (3) captures the fundamental error limit in such scenarios.

III. LINK-LENGTH DEPENDENCE OF ATMOSPHERIC TURBULENCE IN FSO LINKS

In free-space optical (FSO) channels under atmospheric turbulence, the received irradiance often follows a *Gamma-Gamma* (G-G) probability distribution, which models *large-scale* (beam wander) and *small-scale* (scintillation) fluctuations [44], [45], [46], [59]. Let $I \geq 0$ be the normalized irradiance. The G-G distribution has the probability density function (PDF) [44]

$$f_I(i) = \frac{2(\alpha\beta)^{\frac{\alpha+\beta}{2}}}{\Gamma(\alpha)\Gamma(\beta)} i^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}\left(2\sqrt{\alpha\beta}i\right), \quad i > 0, \quad (6)$$

where $K_\nu(\cdot)$ is the modified Bessel function of the second kind, and $\alpha, \beta > 0$ are *shape parameters*. Physically, α is linked to large-scale turbulence (beam wander), and β to small-scale turbulence (faster scintillation cells). Both parameters can be expressed in terms of the *Rytov variance* σ_R^2 that in turn depends on the *link length* L .

A. Rytov Variance Vs. Link Length

In weak-to-moderate turbulence theory, the Rytov variance σ_R^2 (variance of the log-amplitude fluctuations) is often approximated by [44]

$$\sigma_R^2(L) = C C_n^2 k^{\frac{7}{6}} L^{\frac{11}{6}}, \quad (7)$$

where:

- C is a constant of order unity (e.g. 1.23 for plane waves),
- C_n^2 is the refractive-index structure parameter [$\text{m}^{-2/3}$],
- $k = 2\pi/\lambda$ is the wavenumber (λ the wavelength),
- L is the propagation distance (link length).

As L grows, σ_R^2 increases, indicating stronger turbulence effects (sometimes eventually saturating in the strong-turbulence regime).

B. Shape Parameters α and β as Functions of σ_R^2

Once σ_R^2 is known, the shape parameters α, β of the Gamma-Gamma PDF (6) can be obtained via empirical relations [44], [45]. One widely used set of formulas is:

$$\alpha(\sigma_R^2) = \left[\exp\left(\frac{0.49\sigma_R^2}{(1 + 1.11\sqrt{\sigma_R^2})^{\frac{7}{6}}}\right) - 1 \right]^{-1}, \quad (8)$$

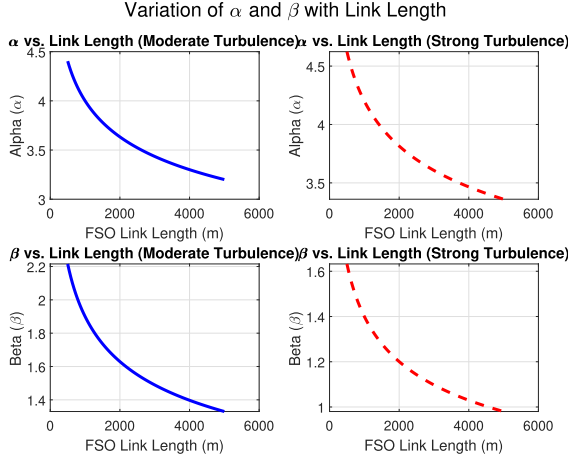


Fig. 1. Variation of α and β with link length over moderate ($C_n^2 = 5 \times 10^{-14}$) and strong ($C_n^2 = 2 \times 10^{-13}$) turbulence.

$$\beta(\sigma_R^2) = \left[\exp \left(\frac{0.51 \sigma_R^2}{(1 + 0.69 \sqrt{\sigma_R^2})^{\frac{5}{6}}} \right) - 1 \right]^{-1}. \quad (9)$$

Hence, for a given link length L , one computes $\sigma_R^2(L)$ using (7), then obtains $\alpha(L) = \alpha(\sigma_R^2(L))$ and $\beta(L) = \beta(\sigma_R^2(L))$ from (9).

1) Comparison of Weak and Strong Turbulences:

- *Under Weak Turbulence* ($\sigma_R^2 \ll 1$): α, β can be large, yielding a less heavy-tailed PDF.
- *Over Strong Turbulence* ($\sigma_R^2 \gg 1$): $\alpha, \beta \rightarrow \mathcal{O}(1)$ or smaller, leading to broader fluctuation distributions and deeper intensity fades.

This models the observed phenomenon that *longer* links (L larger) tend to have stronger scintillation, and thus more severe power fading.

C. Practical Implications for FSO Links

As the link length L grows, one typically sees:

- Higher $\sigma_R^2(L) \propto L^{\frac{11}{6}}$, signifying more severe turbulence;
- $\alpha(L)$ and $\beta(L)$ *decrease*, broadening the Gamma–Gamma PDF (6) and raising the probability of deep fades or outages.

Thus, in many multi-kilometer FSO deployments, Gamma–Gamma turbulence combined with pointing errors can heavily degrade link performance. In this work, we incorporate the dependence $\alpha(L), \beta(L)$ to evaluate the overall *quantum relaying* performance under realistic long-distance scenarios.

The variation of α and β with link length is shown in Fig. 1 for moderate ($C_n^2 = 5 \times 10^{-14}$) and strong ($C_n^2 = 2 \times 10^{-13}$) turbulence. It can be seen from the figure that both parameters decay sharply with link length; making longer range quantum communication almost infeasible due to near-zero values of α and β .

Remark 1: One of the major challenges in FSO quantum communication is the severe impact of atmospheric turbulence and additive noise, which lead to photon loss and decoherence. In our work, we overcome these limitations by proposing a two-hop DF relaying scheme that significantly reduces the

effective turbulence per link. This approach not only improves detection reliability but also mitigates the degradation inherent to long direct FSO channels.

IV. QUANTUM RELAYING OVER FSO LINKS

In this work, we consider a *three-node* quantum communication system, where **Alice** (the transmitter) aims to send quantum-encoded information (via *binary states*) to **Bob** (the ultimate receiver) over a *long* free-space optical channel. To overcome severe turbulence and noise over large distances, we introduce a *relay* node, **Charlie**, situated between Alice and Bob. Consequently, the end-to-end transmission is split into two shorter FSO links:



A. Quantum Relay System Design

The overall system architecture for the proposed quantum relay-assisted FSO communication is illustrated in Fig. 2. The setup comprises three key nodes: Alice (transmitter), Charlie (intermediate relay), and Bob (final receiver). Both classical and quantum domains are employed between Alice–Charlie and Charlie–Bob for encoding, transmission, detection, and re-encoding of information. At the source side, Alice converts classical binary data $\{0, 1\}$ into quantum domain using a classical-to-quantum (C/Q) converter. The binary bit is then encoded into one of two non-orthogonal coherent quantum states (e.g., $|\pm \gamma\rangle$) and transmitted over the atmospheric free-space channel to Charlie. The quantum states are affected by atmospheric turbulence and additive detection noise in the first link (Link #1). Upon reception, Charlie obtains a noisy, mixed quantum state. He performs a quantum measurement (e.g., Helstrom detection) to estimate the transmitted bit. This process involves collapsing the quantum state using a positive operator-valued measure (POVM), thereby converting it back into classical information.¹ This bit is then re-encoded by Charlie into the corresponding quantum state and retransmitted to Bob over the second FSO link (Link #2), following the same process as Alice. In Link #2, the transmitted quantum states again undergo distortion due to turbulence-induced fading and additive noise. At the receiver end, Bob applies quantum decoding and performs a measurement to extract the final estimate of the transmitted bit. This DF relaying strategy divides a long, turbulence-prone FSO link into two shorter segments, improving end-to-end performance in terms of bit detection, especially in scenarios where direct transmission suffers from excessive signal degradation.

Below, we detail the *binary quantum states*, the DF relay operations, and the *Gamma–Gamma* turbulence with *additive noise* for each hop/link.

B. Nodes and Binary Quantum States

1) *Alice (Source/Transmitter):* Alice sends one of two *non-orthogonal* quantum states, typically *coherent states*:

$$|+\alpha\rangle \text{ (bit 0) or } |-\alpha\rangle \text{ (bit 1),}$$

¹We assume uncoded, memoryless transmission for simplicity.

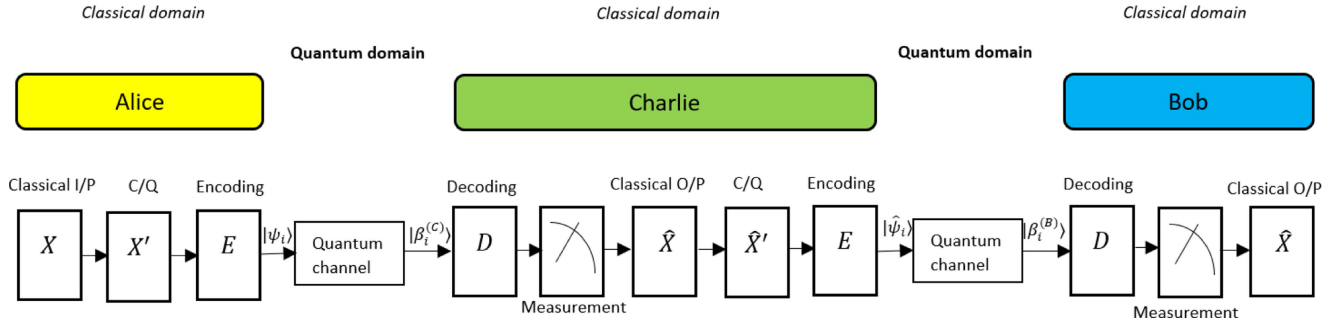


Fig. 2. Quantum FSO relaying.

where $\alpha \in \mathbb{C}$ is the amplitude. Because $\langle +\alpha | -\alpha \rangle \neq 0$, perfect discrimination is impossible; Helstrom's theorem sets a fundamental limit on the minimum error probability [56], [60].

2) *Charlie (Relay)*: After propagating over *Link #1*, the state arrives at Charlie. The *relay* attempts a “decode-and-forward” approach:

- 1) *Estimation*: Charlie performs a *binary quantum measurement* (e.g., near-optimal Helstrom detection for two states) to guess whether $|+\alpha\rangle$ or $|-\alpha\rangle$ was received.
- 2) *Re-encoding*: Based on this guess, Charlie *prepares a new* corresponding coherent state $|+\alpha\rangle$ or $|-\alpha\rangle$, and transmits it towards Bob over *Link #2*.

If the relay misdetects the first link's signal, the bit can become inverted with high confidence. Nevertheless, breaking the single long path into two shorter links can significantly reduce the total turbulence impact.

Remark 2: It is important to note that our DF relay operates without verifying the correctness of its decoding before retransmission to Bob. In other words, the relay may transmit erroneously decoded quantum states, which is an inherent limitation of quantum measurement. This assumption is consistent with the principles of quantum communication, where the security of the message is maintained despite potential errors at intermediate nodes, and no additional trust is placed in the relay.

3) *Bob (Destination/Receiver)*: Finally, Bob receives Charlie's re-transmitted state, once again corrupted by turbulence and noise in *Link #2*. Bob employs a *binary quantum measurement* (e.g., Helstrom or suboptimal threshold detection) to decide which bit was sent [56], [61].

C. FSO Links: Gamma–Gamma Fading and Noise

1) *Link #1 (Alice–Charlie)*: Let I_1 denote the random *irradiance fading* due to atmospheric turbulence. We adopt the *Gamma–Gamma* distribution, which captures large- and small-scale fluctuations. The PDF is [49], [62]

$$f_{I_1}(i) = \frac{2(\alpha_1 \beta_1)^{\frac{\alpha_1 + \beta_1}{2}}}{\Gamma(\alpha_1) \Gamma(\beta_1)} i^{\frac{\alpha_1 + \beta_1}{2} - 1} K_{\alpha_1 - \beta_1}(2\sqrt{\alpha_1 \beta_1} i), \quad i > 0, \quad (10)$$

where α_1, β_1 depend on the *Rytov variance* $\sigma_{R,1}^2$, which in turn depends on *link distance* L_1 . For *weak-to-moderate* turbulence,

$\sigma_{R,1}^2$ is typically approximated via

$$\sigma_{R,1}^2 = C C_n^2 (k^{7/6}) L_1^{11/6}, \quad (11)$$

where C_n^2 is the structure parameter, $k = 2\pi/\lambda$ the wavenumber, and $C \approx 1.23$ for plane-wave conditions [44], [45]. Empirical formulas then map $\sigma_{R,1}^2$ to (α_1, β_1) .

Additionally, each receiver end (here, Charlie) also faces *additive noise*, e.g. shot noise approximated by a Gaussian for moderate flux, or more explicit Poisson counting if the photon number is small.²

2) *Link #2 (Charlie–Bob)*: Similarly, let I_2 denote the fading on the second hop. It too follows *Gamma–Gamma* with parameters (α_2, β_2) , derived from the link distance L_2 and $\sigma_{R,2}^2 \propto L_2^{11/6}$. At Bob's end, we again have additive detection noise. If ρ_{charlie} is the *state* re-encoded by Charlie, the final state at Bob becomes ρ_{charlie} *mixed* with the random amplitude I_2 and the additive noise. Bob performs a *binary quantum measurement* to infer the bit.

Remark 3: Our proposed quantum relaying framework distinguishes itself from both conventional quantum repeaters and classical FSO relays. Unlike traditional quantum repeaters—which rely on complex entanglement distribution, purification, and quantum memories—and classical FSO relays that primarily amplify signals without leveraging quantum security, our two-hop decode-and-forward scheme utilizes binary coherent states combined with optimal quantum measurement techniques to achieve both error suppression and inherent security. By dividing the long FSO link into two shorter segments, our method effectively reduces turbulence-induced photon loss and decoherence, thereby enhancing detection reliability while maintaining resource efficiency. This integrated approach, detailed in the subsequent sections, lays a robust foundation for scalable quantum communication networks.

²We note that additional physical-layer effects such as pointing errors, lens aberrations, and aperture averaging are not explicitly modeled in this work. These phenomena require a separate analytical framework involving spatial jitter, aperture filtering, and detailed optical system modeling. To preserve analytical tractability and focus on the core quantum relay architecture and performance derivations, these effects are left for future investigation as part of extended system-level studies.

D. Overall Impact of Two-Hop Relay

By splitting the *single long* FSO path into $(L_1 + L_2)$, each hop may have *reduced* scintillation (smaller $\sigma_{R,j}^2$) than a straight long path. This can lower the *average* error rate, provided Charlie decodes reliably on the first hop. The *downside* is that an error at Charlie *definitely* flips the bit, so one must carefully analyze the *combined* error over both hops [63]. Typically, the BER of Bob will be:

$$Pe \approx P_{e,1} \times (1 - P_{e,2}) + (1 - P_{e,1}) \times P_{e,2}, \quad (12)$$

where $P_{e,1}$, $P_{e,2}$ are the *respective* minimum-error probabilities for the first and second hops (averaged over I_1 , I_2 , and additive noise).

Remark 4: Equation (12) clearly illustrates that errors incurred at the DF relay propagate to degrade the overall system performance. Even small imperfections in state discrimination at Charlie contribute directly to the cumulative error at Bob, emphasizing that accurate error modeling at the relay is critical. This cascading effect, to be verified by our simulation results, underlines the importance of robust relay performance in maintaining end-to-end fidelity in quantum FSO communication.

Remark 5: A key technical contribution of this paper is the derivation of closed-form expressions for the Helstrom error probability and the associated density operator, which rigorously account for Gamma–Gamma turbulence and additive noise. These analytical results are validated through extensive Monte Carlo simulations, demonstrating excellent agreement across weak, moderate, and strong turbulence regimes.

V. DENSITY OPERATOR CALCULATION

In this section, we will perform the density operator calculation for the considered quantum relaying set-up.

A. Additive Noise

First of all, we need to clarify that we assume that the shot noise is real-valued. Though the optical field can be complex, but the measurement (photocurrent or quadrature amplitude) is a real-valued output; since shot noise is the fluctuation in that real measurement, we model it as a real random variable. For example, in direct detection (typical photodiode), one measures the intensity proportional to the square of the field magnitude producing a real photocurrent. The fundamental fluctuations (shot noise) come from the counting of arriving photons, resulting in a real-valued random process (the photocurrent over time). Further, in homodyne detection (one quadrature), the receiver mixes the signal with a local oscillator and measures one real quadrature of the optical field. The shot noise then appears as a real variation around that measured quadrature. The distribution of the additive noise, which is a Gaussian random variable with mean 0 and variance σ^2 , is given by

$$p_\epsilon(\epsilon) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{\epsilon^2}{2\sigma^2}\right]. \quad (13)$$

B. Mixed States

We label two *pure* coherent states as

$$|\psi_0\rangle \equiv |+\alpha\rangle, \quad |\psi_1\rangle \equiv |-\alpha\rangle,$$

transmitted with prior probabilities p_0 and p_1 , often $p_0 = p_1 = 0.5$ for a symmetric binary scenario.

C. Density Operator

Since the FSO link poses turbulent scenario and quantum measurement offers additive noise, the state transmitted by Bob, i.e., $|\psi_i\rangle$, $i = 0, 1$, will be received at Charlie as

$$\beta_i^{(C)} = \sqrt{I_1}\psi_i + \epsilon_1, \quad (14)$$

where ϵ_1 is the additive noise of the Bob-Charlie link, distributed as (13). Let us assume that I_1 is fixed and ψ_i is also known. Then by using the properties of the Gaussian noise in (14), it can be shown that $\beta_i^{(C)}$ is Gaussian distributed with $\sqrt{I_1}\psi_i$ mean and σ^2 variance. Therefore, the PDF of $\beta_i^{(C)}$ is given by

$$p_{\beta_i^{(C)}}(\beta_i^{(C)}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\beta_i^{(C)} - \sqrt{I_1}\psi_i)^2}{2\sigma^2}\right]. \quad (15)$$

From (15), The density operator for mixed states will be calculated as [22, Eq. (60)]

$$\rho_i^{(C)}(I_1) = \int_{-\infty}^{\infty} p_{\beta_i^{(C)}}(\beta_i^{(C)}) |\beta_i^{(C)}\rangle \langle \beta_i^{(C)}| d\beta_i^{(C)}. \quad (16)$$

A coherent state $|\beta_i^{(C)}\rangle$ (with amplitude $\beta_i^{(C)} \in \mathbb{C}$) can be expanded in the photon-number (Fock) basis $\{|n\rangle\}_{n=0}^{\infty}$ as follows:

$$|\beta_i^{(C)}\rangle = \exp\left(-\frac{1}{2}|\beta_i^{(C)}|^2\right) \sum_{n=0}^{\infty} \frac{(\beta_i^{(C)})^n}{\sqrt{n!}} |n\rangle. \quad (17)$$

Substituting the values of $p_{\beta_i^{(C)}}(\beta_i^{(C)})$ and $\beta_i^{(C)}$ from (15) and (17), respectively, in (16), we get

$$\begin{aligned} \rho_i^{(C)}(I_1) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left[-\frac{(\beta_i^{(C)} - \sqrt{I_1}\psi_i)^2 + 2\sigma^2(\beta_i^{(C)})^2}{2\sigma^2}\right] \\ &\times \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{(\beta_i^{(C)})^{m+n}}{\sqrt{m!}\sqrt{n!}} |m\rangle \langle n| d\beta_i^{(C)}. \end{aligned} \quad (18)$$

From (18), the (m, n) -th element $[\rho_i^{(C)}(I_1)]_{m,n}$ is given by the integral

$$\begin{aligned} [\rho_i^{(C)}(I_1)]_{m,n} &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left[-\frac{(x - \sqrt{I_1}\psi_i)^2 + 2\sigma^2 x^2}{2\sigma^2}\right] \\ &\times \frac{x^{m+n}}{\sqrt{m!}\sqrt{n!}} dx. \end{aligned} \quad (19)$$

A closed-form solution of (19) is obtained in Appendix A as

$$[\rho_i^{(C)}(I_1)]_{m,n} = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{I_1\psi_i^2}{(1+2\sigma^2)}} \sum_{\substack{k=0 \\ k \text{ even}}}^{m+n} \binom{m+n}{k}$$

$$\times \frac{\Gamma\left(\frac{k+1}{2}\right)}{\left(\frac{1+2\sigma^2}{2\sigma^2}\right)^{(k+1)/2}} \left(\frac{\sqrt{I_1}\psi_i}{1+2\sigma^2}\right)^{m+n-k} \frac{1}{\sqrt{m!n!}}. \quad (20)$$

Next, we need to average $[\rho_i^{(C)}(I_1)]_{m,n}$ over Gamma-Gamma distributed I_1 , therefore, We need to evaluate:

$$\mathbb{E}_{I_1} \left[[\rho_i^{(C)}(I_1)]_{m,n} \right] = \int_0^\infty [\rho_i^{(C)}(I_1)]_{m,n} f_{I_1}(i) di, \quad (21)$$

As shown in Appendix B that the final closed-form solution of average density operator $[\rho_i^{(C)}]_{m,n}$ will be

$$[\rho_i^{(C)}]_{m,n} = \frac{1}{\sigma\sqrt{2\pi}} \frac{(\alpha_1\beta_1)^{\frac{\alpha_1+\beta_1}{2}}}{\Gamma(\alpha_1)\Gamma(\beta_1)} \sum_{k=0}^{m+n} \frac{\binom{m+n}{k}}{\sqrt{m!n!}} \quad (22)$$

$$\times \left(\frac{1}{1+2\sigma^2} \right)^{\frac{m+n-k-\alpha_1-\beta_1}{2}} \left(\frac{2\sigma^2}{1+2\sigma^2} \right)^{\frac{k+1}{2}} \quad (23)$$

$$\times \Gamma\left(\frac{k+1}{2}\right) |\psi_i|^{-m-n-\alpha_1-\beta_1+k} \psi_i^{m+n-k} \times G_{1,2}^{2,1} \left(\frac{\alpha_1\beta_1(1+2\sigma^2)}{\psi_i^2} \left| 1 - \frac{m+n-k+\alpha_1+\beta_1}{2}, -\frac{\alpha_1-\beta_1}{2}, -\frac{\alpha_1-\beta_1}{2} \right. \right). \quad (24)$$

VI. HELSTROM DETECTION AND POVM CONSTRUCTION AT CHARLIE

Once we have determined the *impure* density operators $\rho_0^{(C)}$ and $\rho_1^{(C)}$ at Charlie—arising from the two possible binary states (e.g., $|\alpha\rangle$ vs. $|\beta\rangle$) after passing through Gamma-Gamma fading and real amplitude noise—we wish to *distinguish* them with the *minimum average error*. In quantum information theory [64], the *optimal* measurement for such two-state discrimination is given by the *Helstrom measurement*, which yields the *Helstrom bound* on the error probability.

A. Two Mixed States and the Helstrom Operator

We assume two density operators $\rho_0^{(C)}$ and $\rho_1^{(C)}$ with prior probabilities p_0 and p_1 . These operators encapsulate all channel effects (turbulence, pointing errors, additive noise), as detailed in previous sections.

a) *Difference operator*: Helstrom's theory tells us to form

$$\Delta = p_1 \rho_1^{(C)} - p_0 \rho_0^{(C)}.$$

We then diagonalize Δ to find its eigenvalues and eigenvectors.

b) *Constructing the POVM*: If $\Delta = V D V^\dagger$, with $D = \text{diag}(\lambda_0, \lambda_1, \dots)$ the eigenvalue matrix, then define:

$$Q_1 = \sum_{\lambda_k > 0} |\phi_k\rangle \langle \phi_k|, \quad Q_0 = I - Q_1,$$

where $|\phi_k\rangle$ are the eigenvectors of Δ for $\lambda_k > 0$. Physically, outcome “1” occurs in the positive-eigenvalue subspace, outcome “0” in the negative-or-zero subspace. This 2-element POVM $\{Q_0, Q_1\}$ is the *Helstrom measurement* for our two-state discrimination.

B. Helstrom Bound on the Error Probability

The *minimum average error* for deciding between $\rho_0^{(C)}$ and $\rho_1^{(C)}$ with priors p_0, p_1 is given by

$$P_{e,1}^{(\min)} = \frac{1}{2} \left(1 - \text{Tr} |p_1 \rho_1^{(C)} - p_0 \rho_0^{(C)}| \right). \quad (25)$$

Equivalently, if $\Delta = p_1 \rho_1^{(C)} - p_0 \rho_0^{(C)}$, then $\text{Tr} \Delta = \sum_k \lambda_k$, where $\{\lambda_k\}$ are eigenvalues of Δ , then

$$P_{e,1}^{(\min)} = \frac{1}{2} \left(1 - \sum_k \lambda_k \right). \quad (26)$$

This provides the Helstrom (or Helstrom–Holevo) bound for two mixed states at Charlie.

Remark 6: This approach follows the foundational work of Helstrom [60], with generalization to mixed-state discrimination formalized by Holevo [63]. A modern treatment connecting the trace-norm formulation to operational quantum hypothesis testing is given in [64].

VII. DENSITY OPERATOR CALCULATION AND HELSTROM DETECTION AT BOB

Charlie observes a quantum state $\hat{\psi}_i$, which is an estimate of the transmitted state by Alice, i.e., ψ_i . Charlie transmits this estimated quantum state to Bob over the turbulent FSO channel, and the impure state received at Bob will be

$$\beta_i^{(B)} = \sqrt{I_2} \hat{\psi}_i + \epsilon_2, \quad (27)$$

where ϵ_2 is the real-valued Gaussian distributed additive noise of zero mean and σ^2 variance.

The elements of the density operator matrix of Bob, i.e., $\rho_i^{(B)}$, can be obtained similar to Charlie as

$$\begin{aligned} [\rho_i^{(B)}]_{m,n} &= \frac{1}{\sigma\sqrt{2\pi}} \frac{(\alpha_2\beta_2)^{\frac{\alpha_2+\beta_2}{2}}}{\Gamma(\alpha_2)\Gamma(\beta_2)} \sum_{k=0}^{m+n} \frac{\binom{m+n}{k}}{\sqrt{m!n!}} \\ &\times \left(\frac{1}{1+2\sigma^2} \right)^{\frac{m+n-k-\alpha_2-\beta_2}{2}} \left(\frac{2\sigma^2}{1+2\sigma^2} \right)^{\frac{k+1}{2}} \Gamma\left(\frac{k+1}{2}\right) \\ &\times |\hat{\psi}_i|^{-m-n-\alpha_2-\beta_2+k} \hat{\psi}_i^{m+n-k} \\ &\times G_{1,2}^{2,1} \left(\frac{\alpha_2\beta_2(1+2\sigma^2)}{\hat{\psi}_i^2} \left| 1 - \frac{m+n-k+\alpha_2+\beta_2}{2}, -\frac{\alpha_2-\beta_2}{2}, -\frac{\alpha_2-\beta_2}{2} \right. \right). \end{aligned} \quad (28)$$

The *minimum average error* for deciding between $\rho_0^{(B)}$ and $\rho_1^{(B)}$ with priors p_0, p_1 is given by

$$P_{e,2}^{(\min)} = \frac{1}{2} \left(1 - \text{Tr} |p_1 \rho_1^{(B)} - p_0 \rho_0^{(B)}| \right). \quad (29)$$

Equivalently, if $\Delta' = p_1 \rho_1^{(B)} - p_0 \rho_0^{(B)}$ has eigenvalues $\{\lambda'_k\}$, then $\text{Tr} \Delta' = \sum_k \lambda'_k$, leading to

$$P_{e,2}^{(\min)} = \frac{1}{2} \left(1 - \sum_k \lambda'_k \right). \quad (30)$$

Thus one obtains the Helstrom (or Helstrom–Holevo) bound for two mixed states in Bob.

VIII. CLOSED-FORM ERROR RATE FOR HELSTROM DETECTION UNDER AMPLITUDE FADING AND ADDITIVE NOISE

In this section, we derive a closed-form *Helstrom minimum-error probability* for a binary coherent-state detection problem where the received states undergo *fixed turbulence* (I_1) and *real-valued additive noise* (ϵ_1).

The transmitted states experience both amplitude fading and additive noise, leading to the received coherent states:

$$\beta_i^{(C)} = \sqrt{I_1}\psi_i + \epsilon_1. \quad (31)$$

Thus, the two possible received states are:

$$|\psi_0^{(C)}\rangle = |\sqrt{I_1}\gamma + \epsilon_1\rangle, \quad |\psi_1^{(C)}\rangle = |-(\sqrt{I_1}\gamma + \epsilon_1)\rangle.$$

The probability of error in Helstrom's theory depends on the quantum overlap between the two possible received states. It is shown in Appendix E that for two *coherent states* $|+a\rangle$ and $|-a\rangle$, the overlap is given by:

$$\langle a | -a \rangle = \exp(-2|a|^2). \quad (32)$$

In our case, the effective amplitude is:

$$a = \sqrt{I_1}\gamma + \epsilon_1. \quad (33)$$

Thus, the quantum overlap at Charlie is:

$$\langle \psi_0^{(C)} | \psi_1^{(C)} \rangle = \exp(-2|\sqrt{I_1}\gamma + \epsilon_1|^2). \quad (34)$$

Helstrom's minimum error probability for distinguishing two *states* $|\psi_0^{(C)}\rangle$ and $|\psi_1^{(C)}\rangle$ with equal priors ($p_0 = p_1 = 1/2$) is given by:

$$P_{e,1} = \frac{1}{2} \left[1 - \sqrt{1 - |\langle \psi_0^{(C)} | \psi_1^{(C)} \rangle|^2} \right]. \quad (35)$$

Substituting the computed overlap,

$$P_{e,1}(\sqrt{I_1}\gamma + \epsilon_1) = \frac{1}{2} \left[1 - \sqrt{1 - \exp(-4|\sqrt{I_1}\gamma + \epsilon_1|^2)} \right]. \quad (36)$$

Under the assumption that $4|\sqrt{I_1}\gamma + \epsilon_1|^2 \gg 1$, we can use the following relation to get a simplified form of the error rate of Alice, $(1-x)^\zeta \approx 1 - \zeta x$, for $x \ll 1$:

$$P_{e,1}(\sqrt{I_1}\gamma + \epsilon_1) = \frac{1}{4} \exp(-4|\sqrt{I_1}\gamma + \epsilon_1|^2). \quad (37)$$

It is shown in Appendix C that the error rate of Alice conditioned on I_1 will be

$$P_{e,1}(I_1) = \mathbb{E}_{\epsilon_1} \left[P_{e,1}(\sqrt{I_1}\gamma + \epsilon_1) \right] = \frac{e^{-\frac{4I_1\gamma^2}{1+8\sigma^2}}}{4\sqrt{2}\sigma} \sqrt{\frac{1}{4 + \frac{1}{2\sigma^2}}}. \quad (38)$$

In order to find the average error rate of Alice, we need to average (38) over the distribution of I_1 , i.e., $P_{e,1} = \mathbb{E}_{I_1} [P_{e,1}(I_1)]$, which is obtained in Appendix D. From (76), we get

$$P_{e,1} = \frac{1}{4} \sqrt{\frac{1}{1+8\sigma^2}} \cdot \frac{(\alpha_1\beta_1)^{\beta_1}}{\Gamma(\alpha_1)\Gamma(\beta_1)} \left(\frac{4\gamma^2}{1+8\sigma^2} \right)^{-\alpha_1} \times \left[\left(\frac{4\gamma^2}{1+8\sigma^2} \right)^{(\alpha_1-\beta_1)} \Gamma(\alpha_1 - \beta_1) \Gamma(\beta_1) \right]$$

$$\times {}_1F_1 \left(\beta_1, 1 - (\alpha_1 - \beta_1), \frac{\alpha_1\beta_1}{\frac{4\gamma^2}{1+8\sigma^2}} \right) + (\alpha_1\beta_1)^{(\alpha_1-\beta_1)} \Gamma(\beta_1 - \alpha_1) \Gamma(\alpha_1) \times {}_1F_1 \left(\alpha_1, 1 + (\alpha_1 - \beta_1), \frac{\alpha_1\beta_1}{\frac{4\gamma^2}{1+8\sigma^2}} \right) \Big]. \quad (39)$$

The error rate of Bob can be obtained by simple substitution of corresponding variables in (39) as

$$P_{e,2} = \frac{1}{4} \sqrt{\frac{1}{1+8\sigma^2}} \cdot \frac{(\alpha_2\beta_2)^{\beta_2}}{\Gamma(\alpha_2)\Gamma(\beta_2)} \left(\frac{4\gamma^2}{1+8\sigma^2} \right)^{-\alpha_2} \times \left[\left(\frac{4\gamma^2}{1+8\sigma^2} \right)^{(\alpha_2-\beta_2)} \Gamma(\alpha_2 - \beta_2) \Gamma(\beta_2) \right] \times {}_1F_1 \left(\beta_2, 1 - (\alpha_2 - \beta_2), \frac{\alpha_2\beta_2}{\frac{4\gamma^2}{1+8\sigma^2}} \right) + (\alpha_2\beta_2)^{(\alpha_2-\beta_2)} \Gamma(\beta_2 - \alpha_2) \Gamma(\alpha_2) \times {}_1F_1 \left(\alpha_2, 1 + (\alpha_2 - \beta_2), \frac{\alpha_2\beta_2}{\frac{4\gamma^2}{1+8\sigma^2}} \right) \Big]. \quad (40)$$

The end-to-end error rate of the proposed quantum FSO relaying can be obtained in closed-form by substituting (39) and (40) in (12).

Remark 7: Photon loss and decoherence are implicitly modeled in our system through the Gamma–Gamma fading parameter I_i , which captures turbulence-induced amplitude fluctuations. These impairments directly affect the distinguishability of the received coherent quantum states and are incorporated into our analysis via the density operator expressions in (24) and (28), and the corresponding Helstrom error bounds in (39) and (40), which together account for both turbulence and additive noise in each hop. Importantly, the proposed two-hop decode-and-forward relaying strategy serves as a mitigation mechanism against such quantum signal degradation. By breaking the long-range FSO link into two shorter segments, the effective turbulence experienced per hop is significantly reduced, thereby limiting photon loss and decoherence. This design improves the overall reliability and fidelity of quantum communication, as demonstrated through both analytical and simulation results.

Remark 8: While both the *density matrix approach* (employing (24), (25), and (26)) and the approximate *closed-form approach* (given in (39)) yield the same Helstrom minimum error probability, the closed-form expression offers several practical advantages. First, it *eliminates the need for truncation* in the Fock basis, which is required when computing the density operator. This avoids potential *truncation errors* that may arise in high-photon-number regimes. Second, the closed-form solution *directly integrates over turbulence and noise distributions*, leading to an *explicit analytical representation* of the error probability. This provides *greater computational efficiency* compared to the density matrix approach, which requires *eigenvalue decomposition* to compute the trace norm. Additionally, the closed-form formula using *hypergeometric functions* enables *explicit scaling analysis*, allowing one to *predict performance*

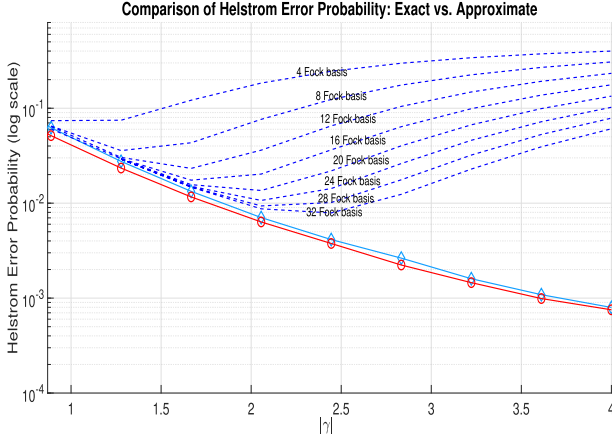


Fig. 3. Comparison of the proposed closed-form approximate error rate (39) —○—, ensemble averaged exact error rate (36) —◇—, and exact error rate via density operator through (24), (25), and (26) —, for Charlie in moderate turbulence scenario, i.e., $\alpha = 4.0$, $\beta = 1.9$, and $\sigma = 1$.

trends based on turbulence strength, additive noise levels, and photon number. Finally, the closed-form solution can be *easily evaluated numerically* using standard mathematical libraries, making it well-suited for *large-scale simulations and performance optimization* in quantum FSO communication systems. Thus, while the density matrix approach offers a more fundamental operator-based perspective, the closed-form solution is often *preferred for practical implementation and large-scale performance evaluation*.

IX. NUMERICAL RESULTS AND DISCUSSION

In this section, we will evaluate the performance of the proposed quantum FSO relaying scheme over turbulence and additive noise.

A. Error Performance of the Proposed Quantum Relaying

Fig. 3 shows error rate of Charlie obtained over Alice-Charlie quantum link by using three methods: i) the exact error rate by using the closed-form expression of the density operator (24) and then using eigenvalue decomposition and computing the trace norm as given in (25) and (26), ii) by doing ensemble averaging of the derived error probability (depending upon overlap of states, i.e., $\langle \psi_0^{(C)} | \psi_1^{(C)} \rangle$) in (36), and iii) plotting the approximate but closed-form analytical error rate given in (39). It can be seen from the figure that derived approximate error rate (plotted directly from (39)) follows the ensemble error rate (obtained by Monte-Carlo simulation from (36)) very closely. Therefore, (39) can be used satisfactorily to predict the performance behavior of the proposed quantum relaying scheme. It can be further seen from the figure that the lengthy method of error rate calculation, i.e., through (24), (25), and (26), leads to extremely slow convergence. Moreover, this method involves hefty calculations due to the requirement to find each element of the density matrix, i.e., $[\rho_0^{(C)}]_{m,n}$ and $[\rho_1^{(C)}]_{m,n}$, separately. Further, the computational complexity grows with the number of Fock basis/states used in computation. Additionally, truncation

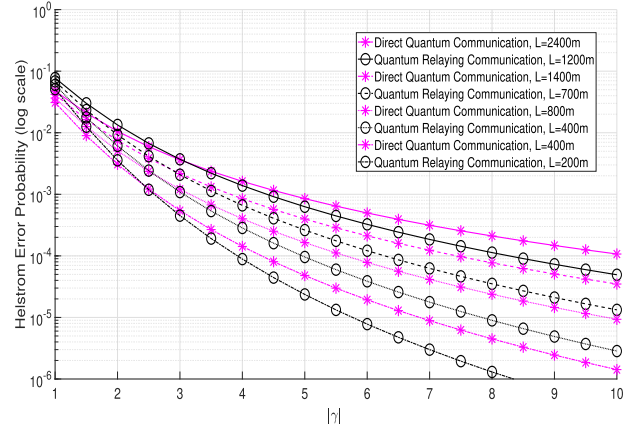


Fig. 4. Comparison of the proposed quantum FSO relaying and direct quantum communication for weak turbulence scenario.

of the Fock basis results in unsatisfactory performance, as can be seen from the figure. Above all, the convergence of the density matrix method is extremely slow; it requires a huge number of Fock basis for satisfactorily predicting the error rate of the receiver, as shown in the figure.

B. Performance Comparison With Direct Quantum Transmission

In Fig. 4, we have shown a comparison of the direct link based quantum communication between Alice and Bob and quantum relaying communication between Alice and Bob via Charlie. We have considered weak turbulence scenario which offers $\alpha = 4.4$ and $\beta = 2.0$ at a link length of 1000 m. It is assumed that Charlie is situated at equal distance from Alice and Bob, which results into significantly improved values of α_i and β_i as compared those of the double length direct link between Alice and Bob. The closed-form Helstrom error is plotted for different values of the field amplitude $|\gamma|$ (which directly depends upon mean photon number) using (12), (39), and (40) for quantum relaying FSO system. For direct quantum communication between Alice and Bob, we employ (39) only by substituting $\alpha_1 = \alpha$ and $\beta_1 = \beta$. The direct link lengths are chosen as follows: $L = 2400$ m, 1400 m, 800 m, 400 m, and the corresponding values of direct link parameters are $\{\alpha_{2400}, \beta_{2400}\} = \{3.3837, 1.5380\}$, $\{\alpha_{1400}, \beta_{1400}\} = \{3.9775, 1.8080\}$, $\{\alpha_{800}, \beta_{800}\} = \{4.7046, 2.1385\}$, $\{\alpha_{400}, \beta_{400}\} = \{5.7921, 2.6328\}$. Since Charlie is kept exactly in between Alice and Bob, both Alice-Charlie and Charlie-Bob link will experience the same turbulence parameters $\{\alpha_{1200}, \beta_{1200}\} = \{4.1658, 1.8935\}$, $\{\alpha_{700}, \beta_{700}\} = \{4.8969, 2.2259\}$, $\{\alpha_{400}, \beta_{400}\} = \{5.7921, 2.6328\}$, $\{\alpha_{200}, \beta_{200}\} = \{7.1309, 3.2413\}$. Further, the value of σ is taken as 0.1. It can be seen from the figure that the proposed quantum relaying scheme performs better than the direct quantum communication, from low-to-moderate values of the field amplitude. At very low field amplitudes, the relaying errors dominate the end-to-end error performance of the quantum FSO relaying scheme, resulting in poor performance than the direct quantum link. Further, it can be observed from

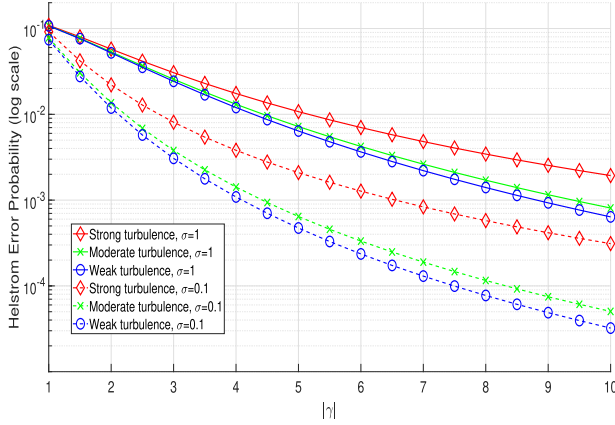


Fig. 5. Performance of the proposed quantum FSO relaying scheme for different turbulence regimes and additive noise values.

TABLE I
GAMMA–GAMMA TURBULENCE PARAMETERS FOR DIFFERENT ATMOSPHERIC CONDITIONS

Turbulence Regime	α	β
Weak	4.4	2.0
Moderate	4.0	1.9
Strong	4.2	1.4

the figure that the performance improvement increases with decreasing link lengths.

Note that the proposed two-way quantum relaying scheme outperforms direct transmission at moderate-to-high field amplitudes under similar turbulence conditions. In real-world scenarios, however, direct links may suffer from obstructions or severe turbulence, rendering them ineffective for quantum transmission. In such cases, our quantum relaying scheme can still deliver satisfactory performance even at low field amplitudes, ensuring robust and reliable quantum communication.

C. Impact of Turbulence Severity and Additive Noise on Relay Performance

In Fig. 5, we present the error performance of the proposed quantum relaying scheme under different turbulence regimes, namely weak, moderate, and strong. It is assumed that both FSO links (Alice–Charlie and Charlie–Bob) experience the same turbulence conditions. The corresponding Gamma–Gamma parameters used to model these regimes are given in Table I.

These parameters reflect different levels of turbulence severity by capturing the combined effects of small-scale and large-scale refractive index fluctuations. More importantly, they are derived from the Rytov variance, which depends on the refractive index structure parameter C_n^2 . This parameter varies with weather conditions and indirectly captures atmospheric factors such as fog, haze, and rain. Thus, the turbulence regimes analyzed in Fig. 5 provide insight into how real-world environmental conditions affect quantum FSO communication.

Additionally, we evaluate the influence of additive detection noise by considering two values of noise variance: $\sigma = 0.1$ (low noise) and $\sigma = 1$ (high noise). It can be seen from the figure that, for a given noise level, the error rate of the quantum

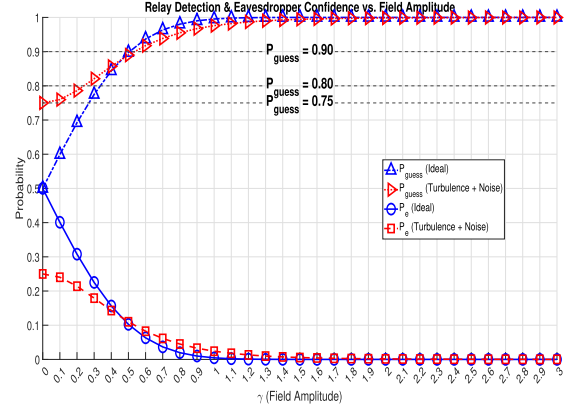


Fig. 6. Relay detection error probability P_e and guessing success probability P_{guess} under ideal and turbulent FSO channels. Thresholds at 0.75, 0.80, and 0.90 are shown to indicate security degradation points.

relaying system increases significantly under strong turbulence conditions compared to weak and moderate regimes. Moreover, as expected, increasing the additive noise level degrades the system's performance across all turbulence cases.

These results confirm that both turbulence and detection noise play critical roles in determining the performance bounds of quantum FSO relaying. Careful design of relay-assisted systems must account for both factors when operating in dynamic or harsh atmospheric environments.

D. Relay Stability Under Noise and Secret Monitoring

In this subsection, we evaluate the operational stability of the quantum relay node (Charlie) under two scenarios: (i) physical channel impairments such as atmospheric turbulence and additive noise, and (ii) potential secret monitoring or malicious behavior by the relay itself. The aim is to characterize how well the relay performs in decoding (robustness), and to what extent it could compromise the system's secrecy if it were untrusted.

1) *Relay Error and Eavesdropper Success Probability:* Let the binary quantum states sent by Alice be $|\psi_0\rangle = |\gamma\rangle$ and $|\psi_1\rangle = |-\gamma\rangle$, where γ denotes the field amplitude. The minimum error probability of distinguishing these states at the relay is given by the Helstrom bound:

$$P_e^{(\text{ideal})} = \frac{1}{2} \left(1 - \sqrt{1 - e^{-4\gamma^2}} \right), \quad (41)$$

and the corresponding guessing success probability of the relay (or a malicious observer) is

$$P_{\text{guess}}^{(\text{ideal})} = 1 - P_e^{(\text{ideal})}. \quad (42)$$

Under turbulent and noisy conditions, we use the closed-form expression derived in (37) to compute the relay error probability $P_e(\gamma, \alpha, \beta, \sigma)$ as a function of Gamma–Gamma fading parameters (α, β) and additive noise variance σ^2 . The guessing probability in this scenario is

$$P_{\text{guess}}^{(\text{real})} = 1 - P_e^{(\text{real})}(\gamma). \quad (43)$$

2) *Stability Under Disturbance and Secret Monitoring:* To assess system stability, we plot both P_e and P_{guess} over a range of field amplitudes $\gamma \in [0, 3]$ for two cases in Fig. 6:

- *Ideal conditions:* No turbulence or noise.
- *Realistic conditions:* Strong turbulence with $(\alpha = 4.2, \beta = 1.4)$ and additive noise with $\sigma = 0.1$.

Threshold lines at $P_{\text{guess}} = 0.75, 0.80$, and 0.90 are included to denote practical secrecy limits. These represent boundaries where the relay begins to gain significant confidence about the transmitted bit and hence compromises secrecy.

Observation 1: Under both ideal and realistic channel conditions, the relay exhibits consistently low error rates for $\gamma > 1.5$, demonstrating detection stability. This confirms that the proposed decode-and-forward relaying strategy is robust with respect to bit detection, even in the presence of strong turbulence and moderate noise.

Observation 2: In the realistic scenario, P_{guess} exceeds key secrecy thresholds (e.g., 0.75 and 0.8) at lower γ values compared to the ideal case. For instance, the $P_{\text{guess}} = 0.80$ level is reached around $\gamma \approx 0.25$ under turbulence, whereas in the ideal case, it only occurs beyond $\gamma \approx 0.35$. This illustrates that turbulence and noise accelerate information leakage at the relay.

Observation 3: Despite the elevated eavesdropping risk with increasing γ , the system still retains a viable operational window (e.g., $0.3 \leq \gamma \leq 0.6$), where relay detection is sufficiently reliable and the guessing probability remains within acceptable security limits.

Observation 4: A non-intuitive crossover is observed between the guessing probability curves of the ideal and realistic scenarios at $\gamma \approx 0.45$. In the mid-range region ($\gamma \in [0.5, 1.5]$), the P_{guess} in the realistic scenario briefly dips below that of the ideal case. This is attributed to noise-induced mixing and fading in the realistic channel, which can temporarily increase quantum state overlap and reduce distinguishability — effectively lowering the relay’s guessing accuracy.

Observation 5: At very low field amplitudes ($\gamma < 0.5$), the relay error probability P_e in the turbulent and noisy scenario is slightly lower than in the ideal case. While this may seem counterintuitive, it is due to stochastic variations in fading and noise that occasionally enhance distinguishability between the received quantum states. This effect is marginal and vanishes at higher amplitudes, where the ideal case naturally achieves better separation.

3) *Interpretation:* The proposed decode-and-forward quantum FSO relaying scheme is shown to be *operationally stable* in turbulent and noisy environments. Even when the relay is considered untrusted, the system can be configured to operate within a regime where both reliability and security are preserved. In ideal conditions, the system naturally maintains high secrecy at low photon levels due to the non-orthogonality of the quantum states. In more adverse conditions, the tradeoff between performance and leakage becomes sharper, but remains manageable with proper parameter selection.

Thus, the scheme performs fairly well under both compromised and ideal scenarios, provided that γ is chosen appropriately to balance between detection accuracy and secrecy leakage.

E. Verification of Analytical Helstrom Error Via Simulation

To validate the analytical expression for the Helstrom error probability derived in Eq. (37), we conduct a Monte Carlo

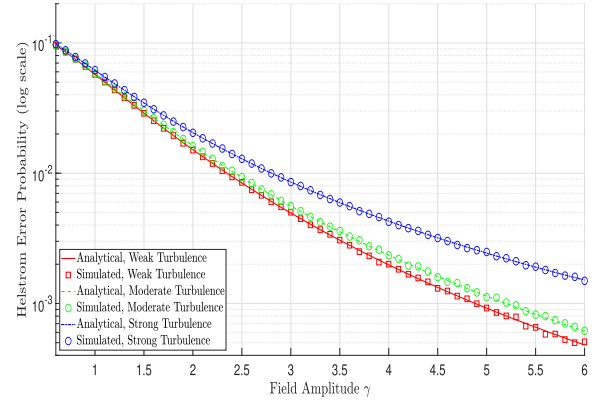


Fig. 7. Comparison of analytical and simulated Helstrom error probabilities at the relay node (Charlie) for weak, moderate, and strong turbulence scenarios.

simulation emulating binary quantum communication over a turbulence-affected FSO link with additive noise, in Fig. 7. The experimental setup simulates the relay node (Charlie), which receives binary coherent states $|\pm\gamma\rangle$ transmitted over a fading and noisy quantum channel. In the simulation, a large number of binary bits (10^6) are randomly generated and mapped to coherent states. Turbulence is modeled using Gamma–Gamma distributed irradiance with parameters (α, β) , while additive noise is modeled as real-valued Gaussian noise with $\sigma = 0.5$. The simulated bit decisions at Charlie are compared with the error probability predicted by the closed-form analytical expression. This validation is performed for weak, moderate, and strong turbulence conditions as depicted in Table 1. In all three scenarios, the empirical error rates observed in the simulation show an almost exact match with the analytical Helstrom error bound across a wide range of field amplitudes γ . This agreement confirms the correctness and applicability of the closed-form error expression in realistic turbulent FSO environments. We emphasize that this verification focuses specifically on the relay node (Charlie), as it is the critical intermediate point in our decode-and-forward protocol. Accurate modeling of the relay’s error rate is essential for assessing the end-to-end performance and security implications of the proposed quantum FSO relaying scheme.

Remark 9: Our proposed framework uniquely integrates classical-to-quantum conversion, quantum state evolution under realistic channel impairments, and optimal quantum measurement theory into a single system model. This comprehensive treatment enhances the originality of the work and lays a robust foundation for scalable quantum communication networks, including potential extensions to multi-hop and satellite-based scenarios.

X. CONCLUSION

This study highlights the significance of relay-based quantum communication in overcoming the challenges posed by long-distance free-space optical (FSO) channels, including signal attenuation, turbulence, and masking effects caused by obstacles. By introducing an intermediate relay node, Charlie, we effectively divide the transmission path into two shorter FSO links, enhancing signal stability and reliability.

Through our proposed quantum relaying method, we analyze the impact of two-hop relaying on error probability, derive the Helstrom error bound, and perform density operator calculations to characterize the quantum relaying setup. Furthermore, the closed-form expressions obtained for key system performance metrics provide a deeper understanding of the feasibility and efficiency of quantum relays in FSO networks. Simulation results confirm that the scheme maintains low error rates under turbulence and noise, while also quantifying information leakage risk using the guessing probability metric.

Compared to other quantum communication models, the proposed coherent-state relay-assisted scheme offers a hardware-efficient alternative to entanglement-based protocols, which typically require quantum memories and complex entangled photon sources [4], [20]. In contrast to continuous-variable QKD systems that rely on Gaussian modulation and homodyne detection [21], our approach uses binary coherent states, enabling simpler detection mechanisms. Additionally, while measurement-device-independent QKD (MDI-QKD) removes trust in the relay by design [24], it introduces considerable implementation complexity. Our model occupies a middle ground—providing practical performance benefits over direct transmission, while still allowing for integration with trust-mitigation protocols.

These findings suggest that the proposed scheme can serve as a foundational building block for quantum network architectures based on FSO links. Future extensions may incorporate quantum error correction, privacy amplification, and entanglement purification to further improve robustness and security in both trusted and untrusted relay environments.

While the present study focuses on a three-node quantum relaying configuration, the proposed analytical framework—based on modular density operator construction and Helstrom error modeling—can be readily extended to scalable multi-hop quantum communication networks. This includes terrestrial as well as hybrid architectures involving ground-to-satellite FSO links, where each hop can be analyzed using the same performance metrics developed in this work.

APPENDIX A

CLOSED-FORM SOLUTION OF INTEGRAL (19)

We need to evaluate:

$$[\rho_i^{(C)}(I_1)]_{m,n} = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp \left[-\frac{(x - \sqrt{I_1}\psi_i)^2 + 2\sigma^2 x^2}{2\sigma^2} \right] \times \frac{x^{m+n}}{\sqrt{m!n!}} dx. \quad (44)$$

The exponent in the integral is:

$$-\frac{(x - \sqrt{I_1}\psi_i)^2 + 2\sigma^2 x^2}{2\sigma^2}. \quad (45)$$

Expanding $(x - \sqrt{I_1}\psi_i)^2$ and substituting into the exponent, we get

$$-\frac{x^2 - 2\sqrt{I_1}\psi_i x + I_1\psi_i^2 + 2\sigma^2 x^2}{2\sigma^2}. \quad (46)$$

A simple rearranging the terms yields:

$$-\frac{(1 + 2\sigma^2)x^2 - 2\sqrt{I_1}\psi_i x + I_1\psi_i^2}{2\sigma^2}. \quad (47)$$

After factoring out $(1 + 2\sigma^2)$ and completing the square, the exponent simplifies to:

$$-\frac{(1 + 2\sigma^2)}{2\sigma^2} \left(x - \frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2} \right)^2 - \frac{I_1\psi_i^2}{(1 + 2\sigma^2)}. \quad (48)$$

Let us define a shifted variable:

$$y = x - \frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2}. \quad (49)$$

Rewriting the integral with help of this substitution:

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{I_1\psi_i^2}{(1+2\sigma^2)}} \int_{-\infty}^{\infty} \left(y + \frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2} \right)^{m+n} e^{-\frac{(1+2\sigma^2)}{2\sigma^2}y^2} dy. \quad (50)$$

Expanding $(y + \frac{\sqrt{I_1}\psi_i}{1+2\sigma^2})^{m+n}$ using the binomial theorem:

$$\left(y + \frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2} \right)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} y^k \left(\frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2} \right)^{m+n-k}, \quad (51)$$

and substituting into (50), we get

$$\sum_{k=0}^{m+n} \binom{m+n}{k} \left(\frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2} \right)^{m+n-k} \int_{-\infty}^{\infty} y^k e^{-\frac{(1+2\sigma^2)}{2\sigma^2}y^2} dy. \quad (52)$$

Employing the standard Gaussian integral formula in (52) [65]:

$$\int_{-\infty}^{\infty} y^k e^{-Ay^2} dy = \begin{cases} \frac{\Gamma(\frac{k+1}{2})}{A^{(k+1)/2}}, & k \text{ is even,} \\ 0, & k \text{ is odd,} \end{cases} \quad (53)$$

we have

$$\int_{-\infty}^{\infty} y^k e^{-\frac{(1+2\sigma^2)}{2\sigma^2}y^2} dy = \begin{cases} \frac{\Gamma(\frac{k+1}{2})}{\left(\frac{1+2\sigma^2}{2\sigma^2}\right)^{(k+1)/2}}, & k \text{ even,} \\ 0, & k \text{ odd.} \end{cases} \quad (54)$$

From (52) and (54), we get the final closed-form expression of $[\rho_i^{(C)}(I_1)]_{m,n}$:

$$[\rho_i^{(C)}(I_1)]_{m,n} = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{I_1\psi_i^2}{(1+2\sigma^2)}} \sum_{\substack{k=0 \\ k \text{ even}}}^{m+n} \binom{m+n}{k} \frac{\Gamma(\frac{k+1}{2})}{\left(\frac{1+2\sigma^2}{2\sigma^2}\right)^{(k+1)/2}} \times \left(\frac{\sqrt{I_1}\psi_i}{1 + 2\sigma^2} \right)^{m+n-k} \frac{1}{\sqrt{m!n!}}. \quad (55)$$

APPENDIX B

AVERAGING $[\rho_i^{(C)}(I_1)]_{m,n}$ OVER GAMMA-GAMMA DISTRIBUTION

To average $[\rho_i^{(C)}(I_1)]_{m,n}$ over Gamma-Gamma distributed I_1 , we proceed as follows:

We need to evaluate:

$$\mathbb{E}_{I_1} [\rho_i^{(C)}(I_1)]_{m,n} = \int_0^\infty [\rho_i^{(C)}(I_1)]_{m,n} f_{I_1}(i) di. \quad (56)$$

Substituting the expressions for $[\rho_i^{(C)}(I_1)]_{m,n}$ and $f_{I_1}(i)$ from (20) and (10), respectively, in (56), the expectation integral becomes:

$$\begin{aligned} \mathbb{E}_{I_1} [\rho_i^{(C)}(I_1)]_{m,n} &= \frac{2(\alpha_1\beta_1)^{\frac{\alpha_1+\beta_1}{2}}}{\sqrt{2\pi}\sigma\Gamma(\alpha_1)\Gamma(\beta_1)} \sum_{\substack{k=0 \\ k \text{ even}}}^{m+n} \binom{m+n}{k} \\ &\times \frac{\Gamma\left(\frac{k+1}{2}\right)}{\left(\frac{1+2\sigma^2}{2}\right)^{(k+1)/2} \sqrt{m!n!}} \\ &\times \underbrace{\int_0^\infty i^{m+n-k+\frac{\alpha_1+\beta_1}{2}-1} e^{-\frac{i\psi_i^2}{(1+2\sigma^2)}} K_{\alpha_1-\beta_1}(2\sqrt{\alpha_1\beta_1}i) di}_{\mathcal{J}}. \end{aligned} \quad (57)$$

Let us use the following substitutions [66]:

$$\begin{aligned} K_\mu(2\sqrt{ax}) &= \frac{1}{2} G_{02}^{20} \left(ax \middle| \begin{matrix} - \\ \mu/2 \end{matrix} \right. \left. -\mu/2 \right), \\ e^{-bx} &= G_{01}^{10} \left(bx \middle| \begin{matrix} - \\ 0 \end{matrix} \right), \end{aligned} \quad (58)$$

in (57), to get:

$$\begin{aligned} \mathcal{J} &= \int_0^\infty I_1^{\nu-1} e^{-bI_1} K_\mu(2\sqrt{aI_1}) dI_1 \\ &= \frac{1}{2} \int_0^\infty x^{\nu-1} G_{01}^{10} \left(bx \middle| \begin{matrix} - \\ 0 \end{matrix} \right) G_{02}^{20} \left(ax \middle| \begin{matrix} - \\ \mu/2 \end{matrix} \right. \left. -\mu/2 \right) dx, \end{aligned} \quad (59)$$

where $\nu = m+n-k + \frac{\alpha_1+\beta_1}{2}$, $b = \frac{\psi_i^2}{(1+2\sigma^2)}$, and $a = \alpha_1\beta_1$. By using [65], [66] in (59), a closed form solution of the integral can be obtained in the form of the Meijer-G function as

$$\mathcal{J} = \frac{b^{-\nu}}{2} G_{12}^{21} \left(\frac{a}{b} \middle| \begin{matrix} 1-\nu \\ \mu/2 \end{matrix} \right. \left. -\mu/2 \right). \quad (60)$$

Substituting (59) in (57), we get (24).

APPENDIX C

AVERAGING (37) OVER ADDITIVE NOISE

Let $I = I_1$ and $\epsilon = \epsilon_1$ for generalization, we need to evaluate:

$$I = \mathbb{E}_\epsilon \left[\frac{1}{4} e^{-4|\sqrt{I}\gamma + \epsilon|^2} \right], \quad (61)$$

where

$$p(\epsilon) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (62)$$

Expanding the squared modulus in (61), we get

$$|\sqrt{I}\gamma + \epsilon|^2 = (\sqrt{I}\gamma + \epsilon)^2 = I\gamma^2 + 2\sqrt{I}\gamma\epsilon + \epsilon^2. \quad (63)$$

Using (63) in (61), the expectation simplifies to the following integral:

$$I = \frac{1}{4\sqrt{2\pi}\sigma} \underbrace{\int_{-\infty}^\infty \exp\left(-4(I\gamma^2 + 2\sqrt{I}\gamma\epsilon + \epsilon^2) - \frac{\epsilon^2}{2\sigma^2}\right) d\epsilon}_{\mathcal{I}}. \quad (64)$$

After some algebra, the exponent in (64) can be written as

$$\begin{aligned} -4(I\gamma^2 + 2\sqrt{I}\gamma\epsilon + \epsilon^2) - \frac{\epsilon^2}{2\sigma^2} &= -4I\gamma^2 - \left(4 + \frac{1}{2\sigma^2}\right) \\ &\times \left(\epsilon + \frac{4\sqrt{I}\gamma}{4 + \frac{1}{2\sigma^2}}\right)^2 + \frac{16I\gamma^2}{4 + \frac{1}{2\sigma^2}}. \end{aligned} \quad (65)$$

Using (65) in (64), we get the revised integral as

$$\begin{aligned} I &= \frac{e^{-4I\gamma^2} e^{\frac{16I\gamma^2}{4 + \frac{1}{2\sigma^2}}}}{4\sqrt{2\pi}\sigma} \\ &\times \underbrace{\int_{-\infty}^\infty \exp\left(-\left(4 + \frac{1}{2\sigma^2}\right)\left(\epsilon + \frac{4\sqrt{I}\gamma}{4 + \frac{1}{2\sigma^2}}\right)^2\right) d\epsilon}_{\mathcal{I}_1}. \end{aligned} \quad (66)$$

Since the integral $\mathcal{I}_1$ is now of the form:

$$\mathcal{I}_1 = \int_{-\infty}^\infty \exp(-A(\epsilon - B)^2) d\epsilon, \quad (67)$$

where $A = 4 + \frac{1}{2\sigma^2}$. Employing the standard Gaussian integral result in (66):

$$\int_{-\infty}^\infty e^{-A(\epsilon-B)^2} d\epsilon = \sqrt{\frac{\pi}{A}}, \quad (68)$$

yields:

$$\begin{aligned} \mathcal{I} &= e^{-4I\gamma^2} e^{\frac{16I\gamma^2}{4 + \frac{1}{2\sigma^2}}} \sqrt{\frac{\pi}{4 + \frac{1}{2\sigma^2}}} \\ &= e^{-4I\gamma^2(1 - \frac{4}{4 + \frac{1}{2\sigma^2}})} \sqrt{\frac{\pi}{4 + \frac{1}{2\sigma^2}}} \\ &= e^{-4I\gamma^2\left(\frac{1}{4 + \frac{1}{2\sigma^2}}\right)} \sqrt{\frac{\pi}{4 + \frac{1}{2\sigma^2}}}. \end{aligned} \quad (69)$$

After some algebra, we get

$$\mathbb{E}_\epsilon \left[e^{-4|\sqrt{I}\gamma + \epsilon|^2} \right] = \frac{e^{-\frac{4I\gamma^2}{1 + 8\sigma^2}}}{4\sqrt{2}\sigma} \sqrt{\frac{1}{4 + \frac{1}{2\sigma^2}}}. \quad (70)$$

APPENDIX D

EXPECTATION OF (38) OVER A GAMMA-GAMMA DISTRIBUTED RANDOM VARIABLE

In this section, we compute the expectation of $\mathcal{I}$ over a Gamma-Gamma distributed random variable I , whose PDF is

given by:

$$f_I(i) = \frac{2(\alpha\beta)^{\frac{\alpha+\beta}{2}}}{\Gamma(\alpha)\Gamma(\beta)} i^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}(2\sqrt{\alpha\beta}i), \quad i > 0. \quad (71)$$

From (38), $\mathcal{L}$ is given by:

$$\mathcal{L} = \frac{e^{-\frac{4i\gamma^2}{1+8\sigma^2}}}{4\sqrt{2}\sigma} \sqrt{\frac{1}{4 + \frac{1}{2\sigma^2}}}. \quad (72)$$

The expected value over I is:

$$\mathbb{E}_I[\mathcal{L}] = \int_0^\infty \frac{e^{-\frac{4i\gamma^2}{1+8\sigma^2}}}{4\sqrt{2}\sigma} \sqrt{\frac{1}{4 + \frac{1}{2\sigma^2}}} f_I(i) di. \quad (73)$$

Substituting the given PDF $f_I(i)$ in (73), we obtain:

$$\mathbb{E}_I[\mathcal{L}] = \frac{1}{4\sqrt{2}\sigma} \sqrt{\frac{1}{4 + \frac{1}{2\sigma^2}}} \frac{2(\alpha\beta)^{\frac{\alpha+\beta}{2}}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty e^{-\frac{4i\gamma^2}{1+8\sigma^2}} i^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}(2\sqrt{\alpha\beta}i) di. \quad (74)$$

The integral encountered in (73) is a Laplace transform of the Gamma-Gamma distribution, which has the known closed-form solution:

$$\begin{aligned} \int_0^\infty x^{v-1} e^{-bx} K_m(2\sqrt{ax}) dx &= \frac{1}{2} b^{-v} (ab)^{-m/2} \\ &\times \left[b^m \Gamma(m) \Gamma\left(-\frac{m}{2} + v\right) {}_1F_1\left(-\frac{m}{2} + v, 1 - m, \frac{a}{b}\right) \right. \\ &\left. + a^m \Gamma(-m) \Gamma\left(\frac{m}{2} + v\right) {}_1F_1\left(\frac{m}{2} + v, 1 + m, \frac{a}{b}\right) \right]. \quad (75) \end{aligned}$$

Using (75) in (74) with the following parameters: $v = \frac{\alpha+\beta}{2}$, $b = \frac{4\gamma^2}{1+8\sigma^2}$, $m = \alpha - \beta$, $a = \alpha\beta$, we get

$$\begin{aligned} \mathbb{E}_I[\mathcal{L}] &= \frac{1}{4} \sqrt{\frac{1}{1+8\sigma^2}} \cdot \frac{(\alpha\beta)^\beta}{\Gamma(\alpha)\Gamma(\beta)} \left(\frac{4\gamma^2}{1+8\sigma^2} \right)^{-\alpha} \\ &\times \left[\left(\frac{4\gamma^2}{1+8\sigma^2} \right)^{(\alpha-\beta)} \Gamma(\alpha - \beta) \Gamma(\beta) \right. \\ &\times {}_1F_1\left(\beta, 1 - (\alpha - \beta), \frac{\alpha\beta}{\frac{4\gamma^2}{1+8\sigma^2}}\right) + (\alpha\beta)^{(\alpha-\beta)} \\ &\times \Gamma(\beta - \alpha) \Gamma(\alpha) {}_1F_1\left(\alpha, 1 + (\alpha - \beta), \frac{\alpha\beta}{\frac{4\gamma^2}{1+8\sigma^2}}\right) \left. \right]. \quad (76) \end{aligned}$$

APPENDIX E PROOF OF COHERENT STATE OVERLAP

Coherent states $|\alpha\rangle$ are defined as eigenstates of the annihilation operator $\hat{a}$:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad (77)$$

where α is a complex number representing the displacement in phase space. The coherent state $|\alpha\rangle$ can be expressed in the Fock

basis as (1):

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (78)$$

To compute the overlap $\langle\alpha| - \alpha\rangle$, we use:

$$\langle\alpha| - \alpha\rangle = e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{(\alpha^*)^n (-\alpha)^n}{n!} \langle n|n\rangle. \quad (79)$$

Since the number states $|n\rangle$ are orthonormal ($\langle n|n\rangle = 1$), the sum simplifies to:

$$\langle\alpha| - \alpha\rangle = e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{(-|\alpha|^2)^n}{n!}. \quad (80)$$

Recognizing the power series expansion of the exponential function:

$$\sum_{n=0}^{\infty} \frac{(-|\alpha|^2)^n}{n!} = e^{-|\alpha|^2}, \quad (81)$$

we obtain:

$$\langle\alpha| - \alpha\rangle = e^{-|\alpha|^2} e^{-|\alpha|^2} = e^{-2|\alpha|^2}. \quad (82)$$

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